

EXPERIMENT 6

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D20A-02

AIM: Creation and Deployment of Smart Contract and Transactions using Solidity and Remix IDE.

THEORY:

1. Smart Contract

A smart contract is a program deployed on a blockchain that automatically executes predefined rules when its functions are called. In this experiment, a Ballot/Voting smart contract is created using Solidity and deployed using Remix IDE.

The Ballot contract allows:

- A chairperson to give voters the right to vote.
- Voters to cast their votes.
- Voters to delegate their votes to another voter.
- The contract to calculate the winning proposal.
- The name of the winning proposal to be displayed

2. require Statement

The require statement is used as a guard condition in Solidity.

It checks whether a particular condition is true before allowing a function to continue executing.

For example:

```
require(msg.sender == chairperson,  
    "Only chairperson can give right to vote.");
```

This ensures that only the chairperson can give someone the right to vote.

If the condition is false:

1. Function execution stops.
2. The transaction is reverted.
3. State changes made during that transaction are reverted.
4. An error message can be displayed.

In the Ballot contract, require is used to check:

- Whether the caller is the chairperson.
- Whether the voter has already voted.
- Whether the voter already has voting rights.

- Whether the voter has the right to vote.
- Whether self-delegation is attempted.
- Whether a valid delegation is being performed.

Therefore, require improves security, correctness and reliability of a smart contract. Pasted text

3. mapping

A mapping is a Solidity data structure that associates a key with a value, similar to a hash table.

Syntax:

```
mapping(keyType => valueType)
```

In the Ballot contract:

```
mapping(address => Voter) public voters;
```

Here:

- address = key
- Voter = value

Therefore, each Ethereum address is associated with its corresponding voter information.

The voter information contains:

- Voting weight
- Whether the voter has voted
- Delegated address
- Proposal voted for

Mappings are efficient for looking up data using a key. However, they do not have a length property and cannot be directly iterated over. Pasted text

4. storage

storage refers to permanent data stored on the blockchain.

For example:

```
Voter storage sender = voters[msg.sender];
```

Here, sender is a reference to the voter information stored in the blockchain.

Changes made through this reference modify the actual stored state.

Storage is:

- Persistent
- Available across transactions
- More expensive because blockchain storage consumes gas

For example, voter information stored in the voters mapping remains available throughout the contract's lifecycle. Pasted text

5. memory

memory is temporary data storage used during a function call.

For example:

```
constructor(bytes32[] memory proposalNames)
```

Here, `proposalNames` is temporarily stored in memory while the constructor executes.

Once the function execution is completed, memory data is discarded.

Compared with storage:

Storage	Memory
Permanent	Temporary
Stored on blockchain	Used during function execution
More expensive	Cheaper
Persists between transactions	Discarded after function execution

6. Why bytes32 instead of string?

The original Ballot contract uses:

```
bytes32 name;
```

instead of:

```
string name;
```

`bytes32` is a fixed-size data type that stores exactly 32 bytes.

Advantages:

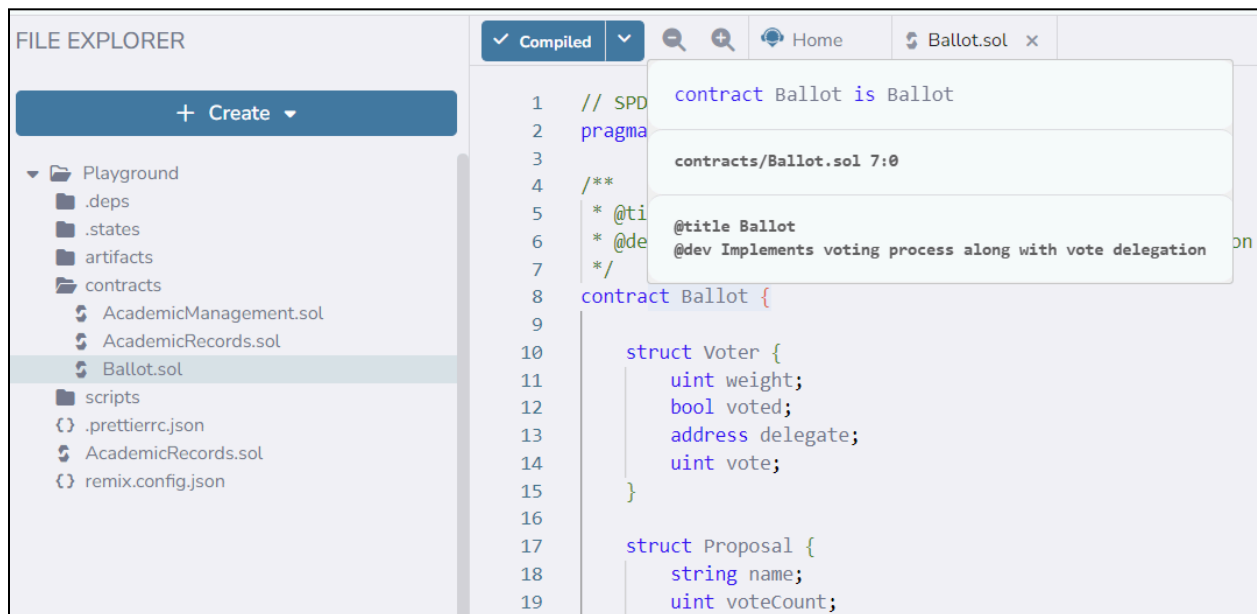
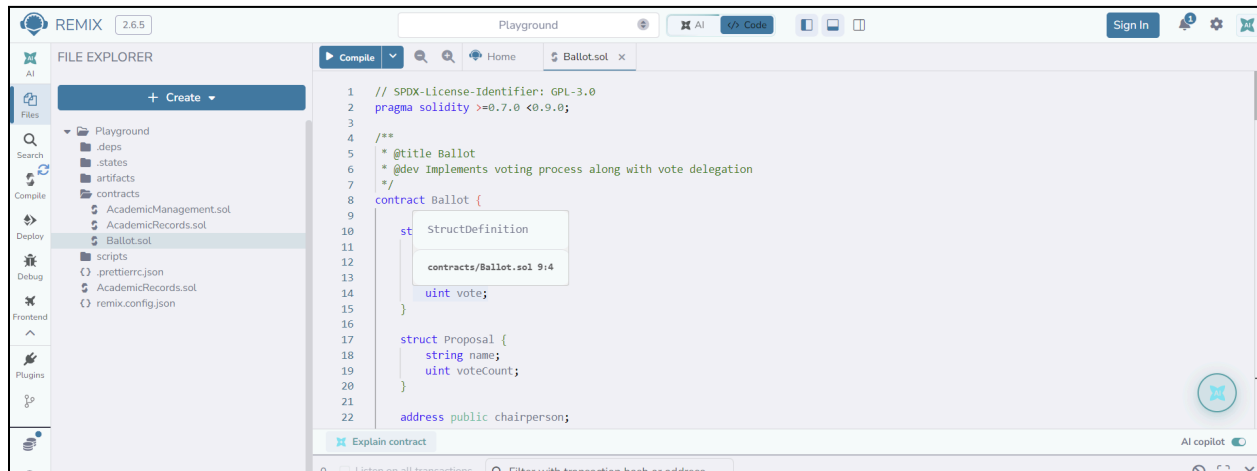
- Efficient storage
- Faster comparisons
- Lower gas usage
- Suitable for short proposal names

However, it limits the proposal name to 32 bytes.

A string is dynamically sized and can store text of variable length. It is more convenient and readable, but can require more complex handling and higher gas usage.

CODE AND OUTPUT:

Creating, compiling and Deploying Ballot.sol on REMIX IDE:



REMIX2.6.5

AI

Files

Search

Compile

Deploy

Debug

Frontend

Plugins

DEPLOY & RUN TRANSACTIONS

Environment

ForkReset

Remix VMOsaka

Account 1100.000 ETH

0x5B3...eddC4

Deploy

Deployed Contracts

Transaction History

Deploy

Remix VM osaka

BallotBallot.sol

proposalNames["Shravanya"]

Value0wei

Gas limitauto0

DEPLOY & RUN TRANSACTIONS

Environment

Remix VM

Osaka

Account 199.999 ETH

0x5B3...eddC4

Deploy

Deployed Contracts

Transaction History

Ballot

Functions

delegate

giveRightToVote

Functions

delegate

giveRightToVote

Compiled

Home

Ballot.sol

1 // SPDX-License-Identifier: GPL-3.0

2 pragma solidity >=0.7.0 <0.9.0;

3

4 /**

5 * @title Ballot

6 * @dev Implements voting process along with vote delegation

7 */

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Deploy

Deployed Contracts 1

Transaction History 3

Calldata Params

Auto-Fill with AI Samples

giveRightToVote (address)

0xAb8483F64d9C6d1EcF9b849Ae677dD3315835

Transact

Calldata Params

Auto-Fill with AI Samples

vote (uint256)

0

Transact

Calldata Params

Auto-Fill with AI Samples

chairperson

Call

Explain contract

0

☐ Listen on all transactions

Filter with transaction hash or address

transaction cost

72984 gas

execution cost

51792 gas

output

0x

decoded input

{
"uint256 proposal": "0"
}

decoded output

{}

logs

[]

raw logs

[]

>

proposals (uint256)

0

Call

Calldata Params

Auto-Fill with AI Samples

0:string: name Shravanya

1:uint256: voteCount 1

decoded input

{
"uint256 ": "0"
}

decoded output

{
"0": "string: name Shravanya",
"1": "uint256: voteCount 1"
}

voters (address)

Call

0xAb8483F64d9C6d1EcF9b849Ae677dD3315835

Calldata Params

Auto-Fill with AI Samples

0:uint256: weight 1

1:bool: voted true

2:address: delegate 0x00

3:uint256: vote 0

giveRightToVote (address)

0x4B20993Bc481177ec7E8f571ceCaE1

Transact

Calldata Params

Auto-Fill with AI Samples

status

1 Transaction mined and execution completed

transaction hash

0xdc8a3d6330b6fbd30ea86dee56d65235dd666446ad5bbb5d669cb862d1af947a

block hash

0xa67f9a56df4502b51a228ed3df03a2ce5580ac56cd6e51cdeec3a6c5d302ed37

Environment Fork Reset

Remix VM Osaka

Account 3 100.000 ETH
0x4B2...C02db

Deploy **Deployed Contracts 1** Transaction History

Calldata Params

Auto-Fill with AI Samples

voters (address) Call

0x4B20993Bc481177ec7E8f571ceCaE8A9e22C02

Calldata Params

Auto-Fill with AI Samples

0:uint256: weight 1
1:bool: voted false
2:address: delegate 0x00
3:uint256: vote 0

Delegated Account 3 to Account 4:

DEPLOY & RUN TRANSACTIONS

Environment Fork Reset

Remix VM Osaka

Account 3 100.000 ETH
0x4B2...C02db

Deploy **Deployed Contracts 1** Transaction History

Functions

Search functions...

delegate (address) Transact

3b7E34aC0F824c42a7cC18A495cabaE

Calldata Params

Auto-Fill with AI Samples

CONCLUSION:

The Ballot/Voting smart contract was successfully created, compiled, and deployed using Solidity and Remix IDE. The experiment demonstrated the practical implementation of voting, delegation, and transaction handling on a blockchain. The use of require statements for security and validation, mapping for associating voter addresses with voter information, and storage and memory for managing data was studied. The difference between bytes32 and string was also understood, particularly in terms of gas efficiency, storage, flexibility, and readability. Overall, the experiment provided practical understanding of the logic, deployment, and execution of a Solidity-based voting smart contract.